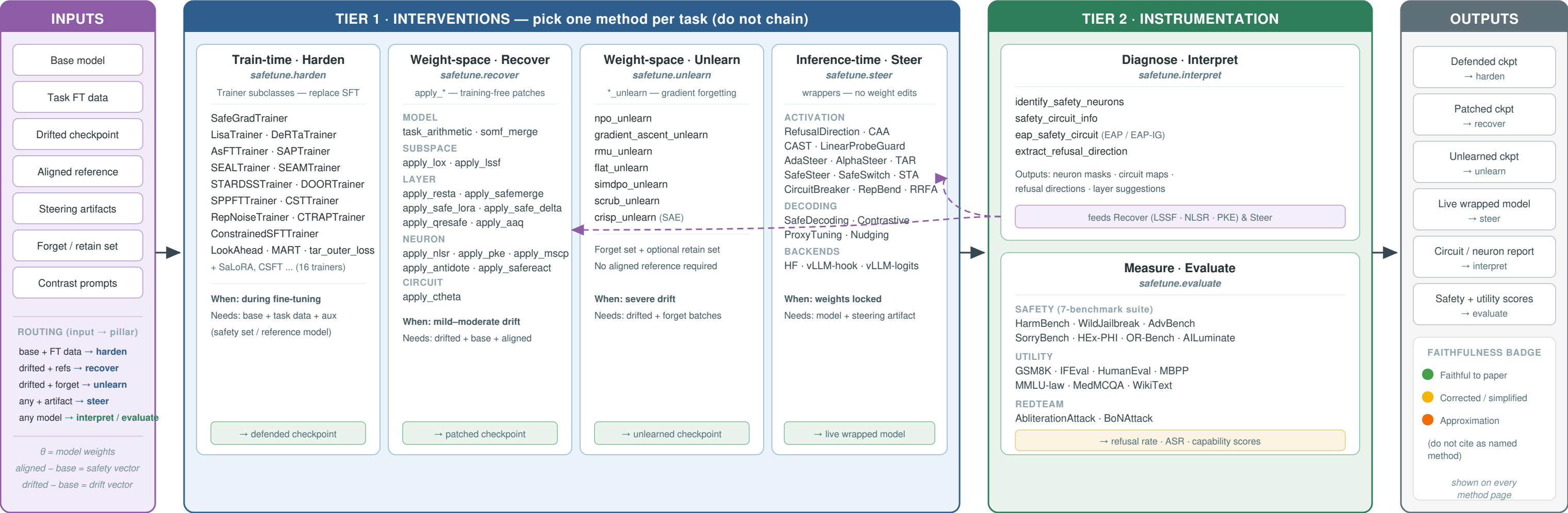


SafeTune — a unified library of alternative LLM-safety methods spanning the fine-tuning lifecycle.

Two tiers: **Interventions** (apply one) and **Instrumentation** (diagnose & measure). Each method carries a faithfulness badge audited against its source paper.



Key finding — paradigm rank is set by drift severity:

Mild drift (e.g. math FT) Recover wins — sub-minute CPU weight arithmetic	Moderate drift Harden or Recover competitive	Severe drift (broad instruction / code) Unlearn wins — gradient-based forgetting (e.g. NPO)
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Figure 1. The SafeTune architecture. Inputs are routed to exactly one Tier-1 intervention — train-time **Harden**, weight-space **Recover** or **Unlearn**, or inference-time **Steer** — each a catalog of independent alternatives rather than a pipeline. Tier-2 **Interpret** locates safety-relevant directions, neurons, and circuits (dashed arrows show how these feed Recover and Steer), while **Evaluate** scores safety and capability on a shared benchmark suite. The recommended paradigm depends on how far the model has drifted from its aligned reference; every method is audited for faithfulness against its source paper.